# Deep Learning for Portfolio Optimization

## Analysis of the paper by Z. Zhang, S. Zohren, and S. Roberts (2020)

**Riccardo Gozzo**

https://github.com/RiccardoGozzo/Deep-Learning-For-Portfolio-Optimization

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# Traditional Portfolio Optimization: Theory vs. Reality

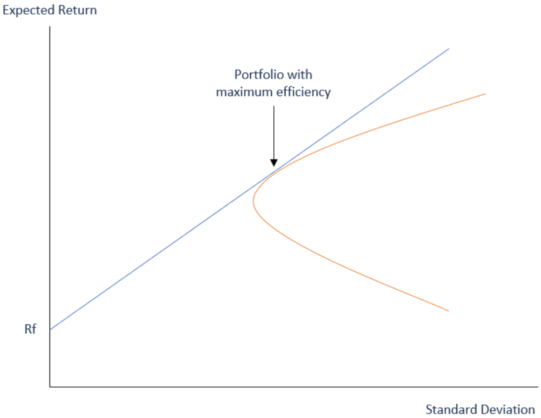



Figure: Theoretical Efficient Frontier

**The Limits of Mean-Variance Optimization**

**Estimation hurdles**: Accurately predicting required future returns and volatility parameters is highly difficult in real-world applications [1, 2]

**Cost erosion**: Management and transaction costs significantly penalize theoretical optimality and final performance.

# Portfolio Benchmarks: MD & MinVar

### Maximum Diversification (MD)

MD maximizes the *Diversification Ratio* ($DR$), extracting the maximum benefit from low asset correlations [3]. The optimal weights $w$ are proportional to the product of inverse covariance and asset volatilities:

$$w = \frac{\Sigma^{-1}\sigma}{\mathbf{1}^T\Sigma^{-1}\sigma} \tag{1}$$

where $\Sigma$ is the covariance matrix and $\sigma$ is the vector of asset-level standard deviations.

### Global Minimum Variance (GMV)

The GMV portfolio represents the point of minimum risk on the Markowitz Efficient Frontier. It is purely risk-based and agnostic to expected returns, making it highly robust against estimation noise:

$$w = \frac{\Sigma^{-1}\mathbf{1}}{\mathbf{1}^T\Sigma^{-1}\mathbf{1}} \tag{2}$$

# Methodological Considerations: The Prediction-Optimization Gap

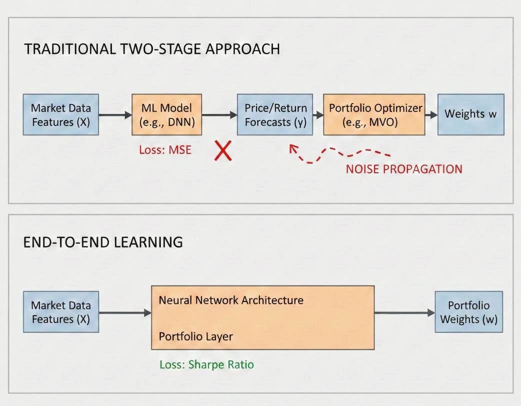



Figure: Traditional vs. End-to-End Learning Workflow.

### Bypassing Forecasting Requirements

Accurate statistical forecasts do not inherently guarantee a maximized Sharpe Ratio.

The study bypasses the estimation of expected returns and covariance matrices, updating portfolio weights directly via model parameters.

This methodology investigates whether internalizing the financial metric into the gradient ascent process mitigates the estimation noise typically propagated by two-stage optimization [4].

# Research Objectives

- **Direct Sharpe Ratio Maximization**: The framework adopts deep learning models to optimize portfolio performance directly, focusing on maximizing the return per unit of risk.
  The methodology bypasses the traditional forecasting of expected returns and covariance matrices, allowing for direct weight updates via model parameters [5, 4].

- **Index-Based Strategy**: The approach utilizes highly liquid market index ETFs to simplify the asset universe and exploit robust cross-asset correlation structures.

# Proposed Architecture and Experimental Protocol (1/2)

- **Validation Protocol:** 5-fold *Expanding Window* cross-validation enforces temporal causality and prevents look-ahead bias.
- **Feature Standardization:** Scaling parameters [6] are derived exclusively from in-sample data to strictly avoid data leakage.
- **Input Processing:** Ingests standardized sliding windows ($T = 50$) to capture sequential dependencies.
- **Recurrent Core:** Single LSTM [7] layer (64 units) regularized via Dropout ($p = 0.2$) and $L_2$ penalty ($10^{-4}$) to mitigate overfitting.

# Proposed Architecture and Experimental Protocol (2/2)

- **Normalization:** Post-recurrent *Layer Normalization* mitigates temporal data leakage and stabilizes learning across distinct market regimes.
- **Allocation Layer:** Zero-initialized Softmax output with ElasticNet $(L_1 + L_2)$ regularization enforces a sparse, diversified distribution.
- **Optimization Dynamics:** Adam optimizer with a conservative learning rate $(\eta = 2 * 10^{-4})$.

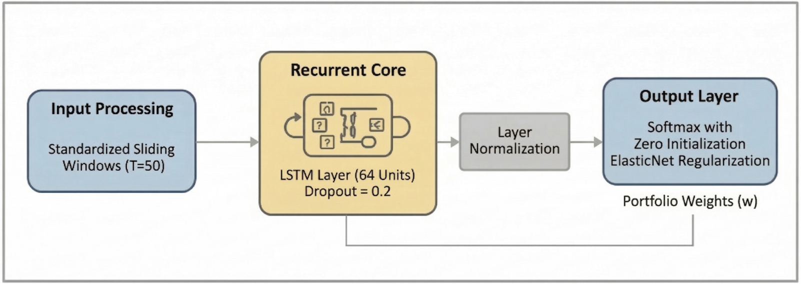



Figure: LSTM architecture

# Objective Function: Net Sharpe Ratio Maximization

## Direct Sharpe Ratio Maximization

The network maximizes the **one-step-ahead** Sharpe Ratio, explicitly incorporating the **risk-free rate** ($r_f$)[1] to target true excess returns. A **many-to-many** LSTM sequence computes turnover by comparing successive allocations ($w_t$ vs $w_{t-1}$), allowing the loss function to internalize rebalancing costs[2]:

$$R_{p,t}^{net} = \underbrace{\left( \sum_{i=1}^{N} w_{i,t} \cdot r_{i,t+1} \right)}_{\text{Gross Return}} - \underbrace{c \cdot \sum_{i=1}^{N} \left| w_{i,t} - w_{i,t-1} \right|}_{\text{Transaction Costs}} - r_{f,t}$$

## The Loss Function

The network minimizes the negative Sharpe Ratio of **net** excess returns over each batch:

$$\mathcal{L}(\theta) = -\frac{\mathbb{E}[R_p^{net}]}{\sigma(R_p^{net})}$$

[1] 3-month T-bill rate
[2] 10 bps

# Experimental Setup: Data and Asset Universe

### Asset Selection and Daily Frequency

The investment universe comprises four highly liquid instruments sampled at a **daily frequency**, representing key asset classes:

- **VTI** (Equity), **AGG** (Bonds), **DBC** (Commodities), **VIX** (Volatility).

### Dataset Features: Prices and Returns

For each asset, the dataset incorporates both **adjusted closing prices** and **returns**. This dual-input approach allows the network to capture both absolute price levels (regime identification) and relative variations (momentum and volatility signals).

*Methodological Note*: Using the VIX index directly (non-tradable) follows the original paper's architecture, despite the simplification regarding tradability.

### Execution Efficiency and Timeframe

High liquidity ensures minimal friction for tradable components. The dataset spans **20 years** (**6 Feb 2006 – 13 Feb 2026**), providing a robust historical backtest.

# Benchmark Selection and Evaluation Regimes

1. **Dual-Regime Stress Test (Out-of-Sample)**
   - **Turbulent (Bad):** *April 2022 – February 2024*. Characterized by high volatility and a lack of clear market drift, testing capital preservation.
   - **Calm (Good):** *February 2024 – February 2026*. Defined by a strong positive drift and low volatility, testing alpha capture efficiency.
2. **Performance Baselines**
   - **Static Naive:** Fixed weights via 5-year look-back.
   - **Dynamic (MinVar/MD):** Convex benchmarks using a rolling-window approach. The covariance matrix is estimated via **Ledoit-Wolf Shrinkage** [8].

| Strategy Proxy | VTI | AGG | DBC | VIX |
|---|---|---|---|---|
| Turbulent (Bad) | 65% | 5% | 20% | 10% |
| Calm (Good) | 60% | 10% | 20% | 10% |

Table: Optimized weights (5-year window).

# Addressing Optimization Instability in Financial Forecasting

## The Instability Problem

Non-convex loss landscapes in finance drive optimizers toward sharp local minima, which cause severe metric fluctuations and poor out-of-sample generalization [9, 10], complicating optimal early-stopping.

## Mitigation Strategies

1. *Heuristic:* Moving averages over epoch metrics to estimate stable stopping regions [11].
2. *ASAM Optimizer:* An adaptive wrapper targeting flat minima to stabilize Sharpe Ratios across regimes [12, 13].

## ASAM Mathematical Formulation

Minimax objective to jointly minimize loss and curvature:

$$\min_{\mathbf{w}} \max_{\|\epsilon\| \leq \rho} L(\mathbf{w} + \epsilon)$$

Adaptive perturbation $\epsilon$ (scaling with parameter magnitude via $T_{\mathbf{w}} = |\mathbf{w}|$):

$$\epsilon = \rho \frac{T_{\mathbf{w}}^2 \nabla L(\mathbf{w})}{\| T_{\mathbf{w}} \nabla L(\mathbf{w}) \|_2}$$

# Extended Feature Engineering

**Hypothesis: Information Enrichment**

Expanding the input space to capture intraday dynamics and structural dependencies beyond closing prices:

- **Volatility Estimators:** Using **Garman-Klass Volatility** (via OHLC) as a more efficient proxy for intraday risk than close-to-close variance.
- **Structural Dependencies:** Adding rolling correlations between the main equity index (VTI) and other assets to signal regime shifts.
- **Trend Over-extension:** Measuring the relative distance of closing prices (VTI, DBC, VIX) from their rolling Simple Moving Averages to capture mean-reversion dynamics.

# Empirical Results: The Signal-to-Noise Dilemma

## The Dimensionality Paradox in Low-SNR Regimes

The **Full Feature Set** underperforms the parsimonious baseline (Prices + Returns).

### Noise Overfitting vs. Generalization

- **Spurious Co-adaptations**: Expanding the feature space in low-SNR environments inherently increases the risk of overfitting on noise [14, 15].
- **Variance Reduction & Stability**: To explicitly avoid sharp minima and target flatter, more generalizable regions [9], we accept a marginal performance drop. The extended features act as a structural regularizer, significantly reducing variance and stabilizing validation.

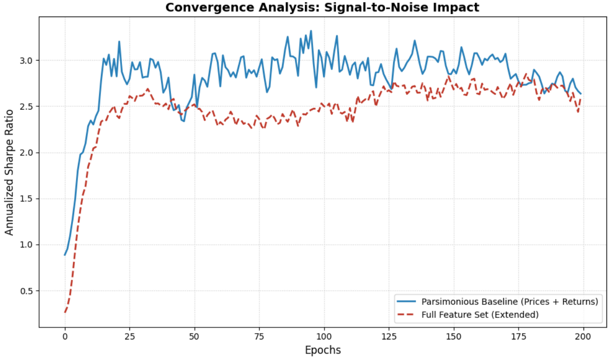



Figure: Performance comparison: Full Feature Set vs. Baseline (Prices + Returns).

# Empirical Results: Performance Across Market Regimes

| Strategy | Ann. Return | Ann. Vol. | Sharpe | Max DD | Avg. Turnover | vs DLS |
|---|---|---|---|---|---|---|
| **High Volatility Regime** *(Apr 2022 - Feb 2024)* | | | | | | |
| **DLS** | **18.43%** | **13.67%** | **1.35** | **-8.04%** | **13.09%** | — |
| Min-Var | 1.02% | 7.10% | 0.14 | -6.39% | 2.34% | $p<0.05$ |
| MD | 3.08% | 8.32% | 0.37 | - 6.41% | 2.10% | $p<0.05$ |
| Naive | 6.93% | 9.87% | 0.70 | -8.03% | 0% | $p<0.05$ |
| **Low Volatility Regime** *(Feb 2024 - Feb 2026)* | | | | | | |
| **DLS** | **51.10%** | **17.57%** | **2.91** | **-12.63%** | **14.76%** | — |
| Min-Var | 9.84% | 6.89% | 1.43 | -4.17% | 1.19% | $p<0.05$ |
| MD | 13.55% | 11.22% | 1.21 | -8.31% | 1.95% | $p<0.05$ |
| Naive | 21.14% | 9.59% | 2.2 | -4.57% | 0% | $p>0.05$ |

Table: Performance comparison, including dynamic rebalancing metrics. **vs DLS**: circular block bootstrap ($B$=10,000, block $\ell = \lfloor 15 \rfloor$ days, pairs resampled jointly).

# Risk Control: Volatility Scaling Dynamics

**Ex-ante Leverage Alignment**
Weights are scaled by a factor $L_{t-1}$ to target a constant risk $\sigma_{tgt}$ ($\sigma_{tgt} = 10\%$) [16, 4, 17]:

$$w_t^{scaled} = w_t \cdot \frac{\sigma_{tgt}}{\hat{\sigma}_{t-1}}, \quad \text{with } \hat{\sigma}_{t-1} = \text{50-day EWMA} \tag{3}$$

**High Vol Regime** (Apr 22 - Feb 24)

| Strategy | Sharpe | Max DD |
|---|---|---|
| **DLS** | **1.09** | **-6.14%** |
| Min-Var | 0.29 | -5.90% |
| MD | 0.60 | -3.83% |
| Naive | 0.89 | -3.76% |

**Low Vol Regime** (Feb 24 - Feb 26)

| Strategy | Sharpe | Max DD |
|---|---|---|
| **DLS** | **2.79** | **-6.19%** |
| Min-Var | 1.95 | -5.57% |
| MD | 2.05 | -4.51% |
| Naive | 2.64 | -2.95% |

**Discussion:** Unlike classical benchmarks, the DLS exhibits a **performance decay** when scaled. This sub-optimal risk-adjustment is driven by high turnover and the greedy behavior of the algorithm.

# Interpretability: Dissecting the "Risk-Seeking" Behavior

**The "Greediness" of the Sharpe Ratio**

Improvements are primarily driven by the **Numerator (Expected Returns)** rather than Volatility reduction.

**Gradient Descent Dynamics**

The analytical gradient with respect to weights $\theta$ highlights this imbalance:

$$\nabla_\theta SR = \frac{1}{\sigma}\nabla_\theta \mu - \frac{\mu}{\sigma^2}\nabla_\theta \sigma \tag{4}$$

- **Adam & Batch Size:** Variance gradient magnitude is intrinsically smaller ($\|\nabla_\theta \sigma\| \ll \|\nabla_\theta \mu\|$). Adam's adaptive normalization becomes noisy and ineffective with small batch sizes [18].
- **Cold Start (Vanishing Penalty):** Early in training, mean returns are near zero ($\mu \approx 0$). This turns off the variance penalty ($\frac{\mu}{\sigma^2} \to 0$), causing the optimizer to greedily seek high yields while ignoring risk.

# Architectural Bias and Theoretical Alignment

### The LSTM as a Low-Pass Filter (Spectral Bias)

RNNs inherently act as spectral low-pass filters [19]:

- **Signal Nature:** Asset returns show persistent trends, while volatility is "bursty" and mean-reverting.
- **Information Transport:** The hidden layer ($h_t$) preferentially propagates trend information, filtering out noisy covariance structures.

### Alignment with Classical Theory

Mirroring deterministic optimization [20], optimal weights are more sensitive to expected returns than covariance. The Deep Learning model replicates this hierarchy, driven primarily by the return vector.

# Optimization Challenges: Mitigation Strategies

## Batch Size Scaling ($B$)

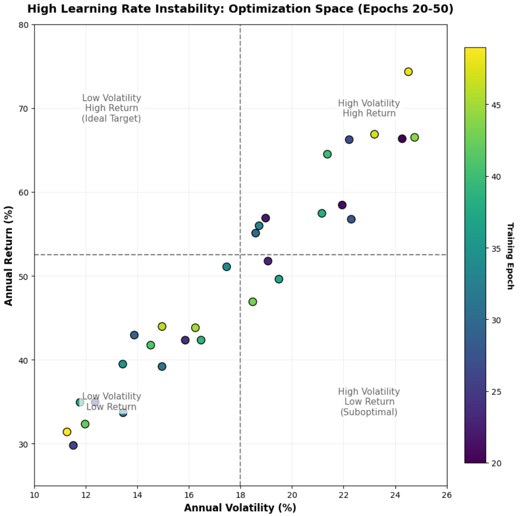
Aims to improve variance gradient precision [18].

- **Constraint**: Data scarcity limits scaling.
- **Result**: Doubling $B$ yields negligible improvements; the signal remains structurally noisy.

## Learning Rate Modulation ($\eta$)

Higher $\eta$ is used to escape local minima and explore the non-convex error surface.

- **Inconsistency**: The model erratically jumps between low-volatility regions and high-variance regimes.
- **No Convergence**: This stochastic oscillation prevents the stability required for live deployment.

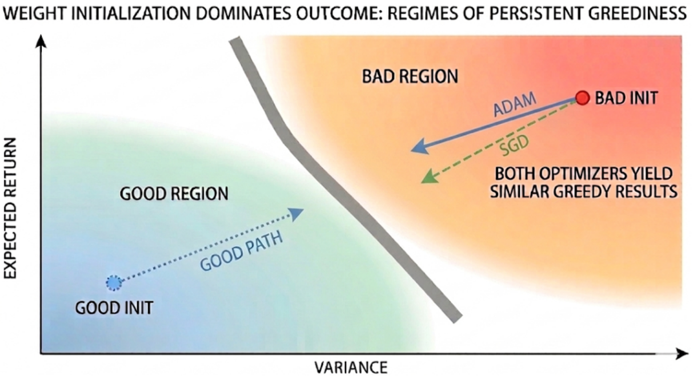



*High LR Instability in Risk-Return Space*

# Optimization Dynamics: The Role of the Optimizer

**Alternative Algorithms**
Optimizers converge to distinct strategies [21]. SGD is tested for its theoretical conservatism and ability to avoid isolated, sharp minima.

**Initialization Bottleneck**
Despite using SGD, results show minimal deviation. The bottleneck lies in **weight initialization** rather than the specific gradient update rule.



# Beyond RNNs: The TSMixer Architecture

To mitigate the *spectral bias* inherent to the recurrent nature of LSTMs, we transition to an **All-MLP architecture** [22, 23].

**TSMixer**[24, 25] replaces recurrence with alternating MLP layers (GELU activations):

- **Time-Mixing**: Extracts temporal patterns across the full window $T$.
- **Feature-Mixing**: Captures cross-sectional correlations between assets.
- **Smart Pooling Head**: Fuses the most recent observation ($x_t$) with the historical average ($\mu_x$) for robust Softmax allocation.

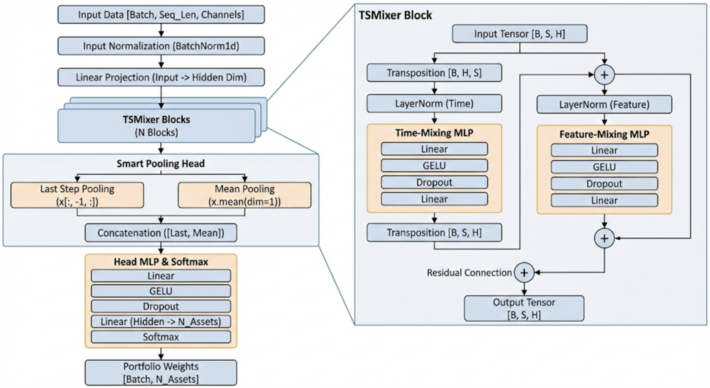



Figure: TSMixer architecture

# Impact of Volatility Scaling Across Regimes

| **Regime / Configuration** | **Ann. Return** | **Ann. Vol.** | **Sharpe** | **Max DD** | **Avg. Turnover** |
|---|---|---|---|---|---|
| **High Volatility Regime** *(Apr 2022 - Feb 2024)* | | | | | |
| TSMixer | 16.64% | 18.46% | 0.90 | -12.64% | 15.07% |
| TSMixer (Scaled 10%) | **6.71%** | **10.83%** | **0.62** | **-6.54%** | **10.89%** |
| **Low Volatility Regime** *(Feb 2024 - Feb 2026)* | | | | | |
| TSMixer | 50.72% | 18.65% | 2.72 | -7.85% | 14.52% |
| TSMixer (Scaled 10%) | **24.57 %** | **10.00%** | **2.46** | **-6.19%** | **9.22%** |

Table: Performance including dynamic rebalancing metrics.

# Optimization Strategy: Transitioning to Linear Utility Loss

A **linear** formulation ($L = -\mu + \lambda\sigma^2$) eliminates the Sharpe Ratio's dynamic coupling, mimicking Mean-Variance utility. This allows for explicit modeling of **risk aversion** via the $\lambda$ parameter.

**Additive Risk Penalty:** The variance gradient is structurally independent of the return magnitude ($\mu$), preventing the risk penalty from vanishing during training "cold starts" ($\mu \approx 0$).

| Configuration | Ann. Return | Ann. Vol. | Sharpe | Max DD | Avg. Turn. | vs Benchmarks |
|---|---|---|---|---|---|---|
| **High Vol Regime** *(Apr 2022 - Feb 2024)* | | | | | | |
| New Loss | 14.96% | 11.43% | 1.31 | -8.30% | 11.88% | $p$<0.05 (all) |
| Scaled 10% | **13.46%** | **10.46%** | **1.29** | **-7.02%** | **9.39%** | - |
| **Low Vol Regime** *(Feb 2024 - Feb 2026)* | | | | | | |
| New Loss | 28.14% | 9.70% | 2.90 | -6.59% | 8.74% | $p$<0.05 (all) |
| Scaled 10% | **31.37%** | **10.67%** | **2.94** | **-6.38%** | **9.24%** | - |

Table: Performance including dynamic rebalancing metrics.

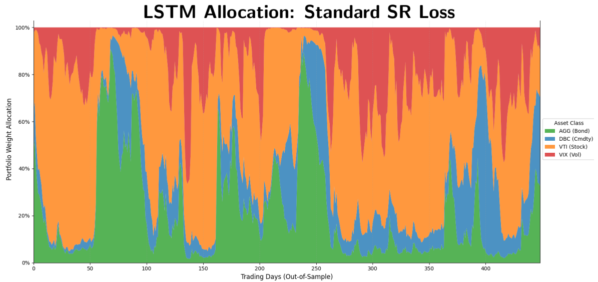
### Turnover Sensitivity

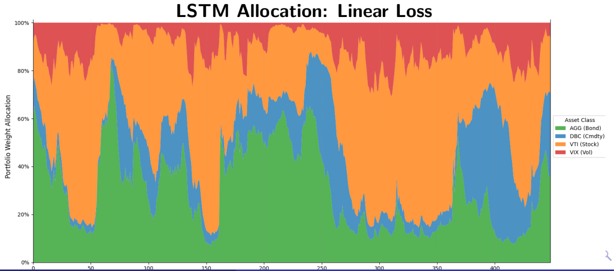
Daily weight turnover depends heavily on the objective function:

- **Standard SR Loss**: 14.76%
- **Linear Loss**: 8.74%

### Allocation Dynamics

- **Standard SR Loss**: Aggressive market timing with abrupt oscillations between Equities (VTI) and Volatility (VIX).
- **Linear Loss**: Smoother transitions, reduced timing aggression, and a structural rebalancing towards Bonds (AGG) for variance control.

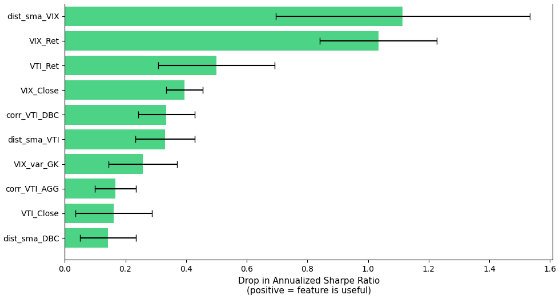


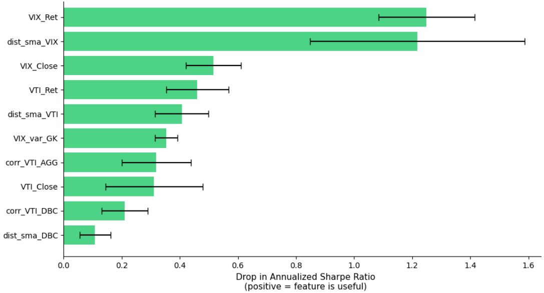
# Interpretability: Permutation Feature Importance

**Permutation Importance** isolates the predictive value of each input by randomly shuffling its sequence across samples and measuring the resulting *drop in the annualized Sharpe Ratio*.

The network extracts the **same foundational signals** regardless of the objective. The loss function (SR vs. Linear Utility) merely dictates how **aggressively** these signals are translated into final allocation weights.




**Standard SR Loss**




**Linear Utility Loss**

# Concluding Remarks and Key Insights

**1. Enhancing Generalization and Structural Stability**

To mitigate overfitting in highly stochastic financial environments, the framework prioritizes robustness over peak performance. Integrating an adaptive optimization wrapper (ASAM) to navigate toward flat minima, coupled with feature augmentation (OHLC, SMAs), consciously trades a marginal drop in absolute metrics for a significant reduction in result variance, ensuring stable and generalizable Out-of-Sample allocations.

**2. Mitigating the "Momentum Trap"**

Standard Sharpe maximization exhibits a tendency to prioritize the first moment (Return). Adopting a **Linear Utility Loss** function provides an additive risk penalty that does not vanish during early training, promoting a more risk-aware allocation in the observed tests.

# Possible Research Directions

**Low-Frequency Scalability**

Adapting the architecture for institutional constraints via **monthly rebalancing**. The network predicts a stable allocation vector for a 21-day window, maximizing the monthly Sharpe Ratio while minimizing turnover. To ensure a realistic and physically tradable strategy, the **VIX index is excluded** from the investable universe.

**Regime-Adaptive Mixture of Experts (MoE)**

Developing a **Mixture of Experts (MoE)** architecture where distinct sub-networks specialize in specific market regimes (e.g., Bull, Bear, High Volatility).

# Bibliography I


[1] H. Markowitz, "Portfolio selection," *The Journal of Finance*, vol. 7, no. 1, pp. 77–91, 1952.

[2] R. O. Michaud, "The markowitz optimization enigma: Is 'optimized' optimal?," *Financial Analysts Journal*, vol. 45, no. 1, pp. 31–42, 1989.

[3] Y. Choueifaty and Y. Coignard, "Toward maximum diversification," *The Journal of Portfolio Management*, vol. 35, pp. 40–51, 2008.

[4] Z. Zhang, S. Zohren, and S. Roberts, "Deep reinforcement learning for trading," 2019.

[5] J. Moody and M. Saffell, "Learning to trade via direct reinforcement," *IEEE Transactions on Neural Networks*, vol. 12, no. 4, pp. 875–889, 2001.

[6] M. Abadi, P. Barham, J. Chen, Z. Chen, A. Davis, J. Dean, M. Devin, S. Ghemawat, G. Irving, M. Isard, M. Kudlur, J. Levenberg, R. Monga, S. Moore, D. G. Murray, B. Steiner, P. Tucker, V. Vasudevan, P. Warden, M. Wicke, Y. Yu, and X. Zheng, "Tensorflow: A system for large-scale machine learning," 2016.

[7] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Computation*, vol. 9, no. 8, pp. 1735–1780, 1997.

[8] W. M. Ledoit, Olivier, "Honey, i shrunk the sample covariance matrix," 2003.

[9] S. Hochreiter and J. Schmidhuber, "Flat minima," *Neural Computation*, vol. 9, no. 1, pp. 1–42, 1997.

[10] N. S. Keskar, D. Mudigere, J. Nocedal, M. Smelyanskiy, and P. T. P. Tang, "On large-batch training for deep learning: Generalization gap and sharp minima," 2017.

[11] N. T. Nguyet, N. B. Chan, and P. T. Anh, "Algoxpert alpha research framework: A rigorous is–wfa–oos protocol for mitigating overfitting in quantitative strategies," technical report, AlgoXpert Lab, 2026.

# Bibliography II


[12] R. Ilbert, A. Odonnat, V. Feofanov, A. Virmaux, G. Paolo, T. Palpanas, and I. Redko, "Samformer: Unlocking the potential of transformers in time series forecasting with sharpness-aware minimization and channel-wise attention," *arXiv preprint arXiv:2402.10198*, 2024.

[13] D. Kahra and V. Kahra, "Vsf-net: A variational encoder and sharpness-aware attention framework for robust forecasting of non-stationary financial time series," *International Journal of Information Technology*, 2025.

[14] L. Zhang and L. Hua, "Major issues in high-frequency financial data analysis: A survey of solutions," *Mathematics*, vol. 13, no. 3, 2025.

[15] F. Moreno-Pino and S. Zohren, "Deepvol: Volatility forecasting from high-frequency data with dilated causal convolutions," *Quantitative Finance*, vol. 24, no. 8, pp. 1105–1127, 2024.

[16] A. MOREIRA and T. MUIR, "Volatility-managed portfolios," *The Journal of Finance*, vol. 72, no. 4, pp. 1611–1643, 2017.

[17] B. Lim, S. Zohren, and S. Roberts, "Enhancing time series momentum strategies using deep neural networks," *arXiv preprint arXiv:1904.04912*, 2019.

[18] M. Lopez de Prado, *Advances in Financial Machine Learning*.
Hoboken, NJ: John Wiley & Sons, 2018.

[19] N. Rahaman, A. Baratin, D. Arpit, F. Draxler, M. Lin, F. Hamprecht, Y. Bengio, and A. Courville, "On the spectral bias of neural networks," in *International conference on machine learning*, pp. 5301–5310, PMLR, 2019.

[20] V. K. Chopra and W. T. Ziemba, "The effect of errors in means, variances, and covariances on optimal portfolio choice," *Journal of Portfolio Management*, vol. 19, no. 2, pp. 6–11, 1993.

# Bibliography III


[21] F. V. Cortesi, G. Iannone, G. Crippa, T. Poggio, and P. Beneventano, "Same error, different function: The optimizer as an implicit prior in financial time series," 2026.

[22] A. Zeng, M. Chen, L. Zhang, and Q. Xu, "Are transformers effective for time series forecasting?," in *Proceedings of the AAAI conference on artificial intelligence*, vol. 37, pp. 11121–11128, 2023.

[23] A. Das, W. Kong, A. Leach, S. Mathur, R. Sen, and R. Yu, "Long-term forecasting with tide: Time-series dense encoder," *arXiv preprint arXiv:2304.08424*, 2023.

[24] S.-A. Chen, C.-L. Li, N. Yoder, S. O. Arik, and T. Pfister, "Tsmixer: An all-mlp architecture for time series forecasting," *arXiv preprint arXiv:2303.06053*, 2023.

[25] V. Ekambaram, A. Jati, N. Nguyen, P. Sinthong, and J. Kalagnanam, "Tsmixer: Lightweight mlp-mixer model for multivariate time series forecasting," in *Proceedings of the 29th ACM SIGKDD conference on knowledge discovery and data mining*, pp. 459–469, 2023.